

Lecture 01

Computer Systems

Chapter lecture — 12 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XI · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — گیارہویں جماعت

12 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XI

How to use this BIEK Computer Science lecture

Each PDF is a classroom lecture from the BIEK / Sindh Computer Science textbooks (New Sindh Curriculum, STBB Grade XI–XII). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before looking at the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر بی آئی کے / سندھ نصاب کی کتاب کے درس کے مطابق ہے۔ سبق پڑھیں، مثال حل کریں، پھر چیک یورسیلف کے سوالات کے جواب اپنے نوٹ بک میں لکھیں۔ کتاب کے بعد اضافی بورڈ پریکٹس ہے۔ ★ گولڈن ٹاپکس امتحان میں زیادہ پوچھے جاتے ہیں۔

Lectures in this pack

Chapter 1: Computer Systems. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems

L02. 1.1.3 Analog and Digital Signals

L03. 1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

L04. 1.1.7 Basic Logic Gates (AND, OR, NOT)

L05. 1.1.7 Universal & Advanced Logic Gates

L06. 1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams

L07. 1.1.13 Karnaugh Maps (K-Maps) Simplification

L08. LogiSim Simulation Tool Guide

L09. 1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)

L10. 1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies

L11. 1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model

L12. 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

Computer Systems

This chapter shows you how a computer thinks in 0s and 1s, how logic gates turn those bits into decisions, how software is planned and built step by step, and how data travels across a network. These three ideas — digital logic, software development and communication models — are the foundation of everything else in Computer Science.

اس باب میں آپ سیکھیں گے کہ کمپیوٹر صفر اور ایک کی زبان میں کیسے کام کرتا ہے، لاجک گیٹس فیصلے کیسے کرتے ہیں، سافٹ ویئر مرحلہ وار کیسے بنایا جاتا ہے، اور نیٹ ورک پر ڈیٹا ایک جگہ سے دوسری جگہ کیسے پہنچتا ہے۔

What you will learn

- Tell the difference between discrete and continuous quantities, and between analog and digital signals, with examples.
- Use the Boolean operations AND, OR and NOT, and apply the order of precedence.
- Draw the symbol, write the expression and give the truth table of AND, OR, NOT, NAND, NOR, XOR and XNOR gates.
- Simplify Boolean expressions using 2-variable and 3-variable Karnaugh maps.
- Describe the six phases of the SDLC and compare the Waterfall and Agile models using case studies.
- State the advantages of digital systems.
- Build a complete truth table for any expression with 2 or 3 inputs.
- Write minterms (SOP) and maxterms (POS) from a truth table, and draw a logic diagram from an expression.
- Build and test a simple circuit in LogiSim Evolution.
- Explain the seven layers of the OSI model and map them to the four layers of the TCP/IP model.

1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems

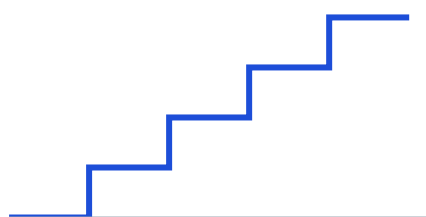
LEARN IT

Everything around you that can be measured or counted is a **quantity**. Quantities come in two kinds.

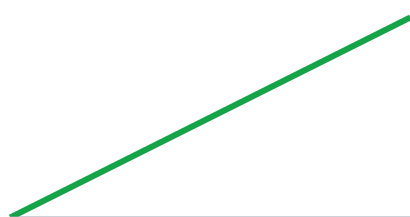
Discrete quantities are separate, countable items with specific values. You can count them one by one: 1, 2, 3 ... There is nothing "in between". You cannot have 2.5 students in a classroom.

Continuous quantities can take *any* value within a range. They flow smoothly without breaks. Water can be 30 °C, 30.5 °C, 30.51 °C, and so on — there is always another value in between.

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Discrete (Stairs)



Continuous (Ramp)

Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

REMEMBER IT

Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask yourself: "Do I *count* it or *measure* it?" Count → discrete. Measure → continuous.

COMMON MISTAKE

Students sometimes say "money is continuous because amounts are large". No — rupees and paise are counted in fixed units, so money is discrete. Size does not matter; what matters is whether values in between are possible.

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔ مسلسل (Continuous) مقداریں کسی حد کے اندر کوئی بھی قیمت لے سکتی ہیں، جیسے پانی کا درجہ حرارت۔ سیڑھیاں منفصل ڈیٹا کی مثال ہیں اور ڈھلوان (ریمپ) مسلسل ڈیٹا کی۔

CHECK YOURSELF

1. Is the number of pages in a book discrete or continuous?
2. Is your height discrete or continuous?
3. In the stairs/ramp analogy, which one shows continuous data?

Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.

LEARN IT

A **digital system** is an electronic system that uses only two values: **0** and **1**. These two values are called **binary digits**, or **bits** for short.

- **0** represents OFF, LOW, or FALSE.
- **1** represents ON, HIGH, or TRUE.

Why only two values? Because it is easy for an electronic circuit to tell "voltage present" from "no voltage". Every number, letter, photo and song in a computer is finally stored as a long pattern of these bits.

Advantages of digital systems

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Examples of digital systems

- **Computers:** process data as 0s and 1s.
- **Smartphones:** store photos and music as binary data.
- **Digital cameras:** capture images as binary code.

WORKED EXAMPLE

Why is copying digital data "accurate"?

Step 1: A song is stored as bits, for example 10110010...

Step 2: When you copy it to a USB, the computer copies each bit exactly: 1 stays 1, 0 stays 0.

Step 3: Even after 100 copies the bits are identical, so the song sounds the same. An old cassette tape (analog) loses a little quality every time it is copied.

REMEMBER IT

Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

اردو میں سمجھیں

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔ 0 کا مطلب بند، کم یا غلط ہے اور 1 کا مطلب چالو، زیادہ یا درست ہے۔ ڈیجیٹل نظام قابلِ بھروسہ، درست، بنانے میں آسان اور سافٹ ویئر سے قابو میں رہنے والے ہوتے ہیں۔

CHECK YOURSELF

1. What does a 1 represent in a digital system?
2. Give two advantages of digital systems.

Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A discrete quantity is:

- a. Counted in separate steps
- b. Any value on a smooth range
- c. Always analog
- d. A sine wave

MCQ 2. Temperature of tea is:

- a. Continuous
- b. Discrete
- c. Only Boolean
- d. A gate

MCQ 3. Number of students in a class is:

- a. Discrete
- b. Continuous
- c. Analog voltage
- d. A ramp

MCQ 4. The stairs vs ramp analogy: stairs are:

- a. Discrete
- b. Continuous
- c. TCP
- d. Agile

MCQ 5. Digital systems use:

- a. 0 and 1
- b. Every real number
- c. Only sine waves
- d. Paper only

MCQ 6. A bit is:

- a. A binary digit, 0 or 1
- b. Eight bytes
- c. A logic gate
- d. A database

Extra short questions

1. Differentiate discrete and continuous with one example each.
2. Why do computers prefer digital values?
3. What is a digital system?
4. Give two advantages of digital systems.

Extra long question

1. With a labelled stairs/ramp sketch, explain discrete vs continuous and link both to analog vs digital signals.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Discrete: countable steps (cars in a park). Continuous: any value in a range (speed of a car).
2. Bits 0/1 can be restored after noise; analog waves are bent by interference.
3. An electronic system that stores and processes only two values, 0 and 1.
4. Reliable (noise resistant), accurate copies, easy to design (two states), programmable.

Long

1. Stairs = discrete/digital (fixed levels). Ramp = continuous/analog (smooth). Computers convert analog measurements into bits, process bits, then convert back if a human must see the result.

1.1.3 Analog and Digital Signals ★ Golden

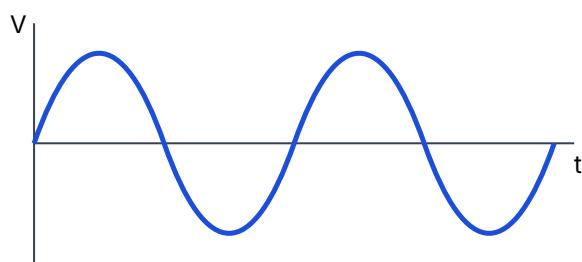
LEARN IT

A **signal** is a changing voltage (or other quantity) that carries information.

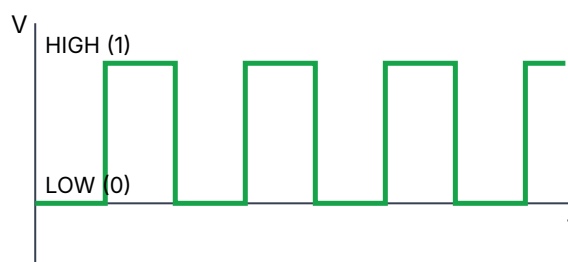
Analog signals are continuous signals that change smoothly over time, like a **sine wave**.

Examples: the human voice, a traditional mercury thermometer.

Digital signals have only two levels: **HIGH (1)** and **LOW (0)**. They jump instantly between these values, which creates a **square wave**. Example: computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.

WORKED EXAMPLE

Why is digital more resistant to noise?

Step 1: Suppose a wire carries 5 V for HIGH and 0 V for LOW.

Step 2: Noise adds a small error, so 5 V arrives as 4.6 V and 0 V arrives as 0.3 V.

Step 3: The receiver only asks "Is it closer to HIGH or LOW?" 4.6 V is still read as 1 and 0.3 V is still read as 0. The data is perfect.

Step 4: In an analog signal, every exact value carries meaning, so 4.6 V instead of 5 V really changes the sound or reading. The noise cannot be removed.

REMEMBER IT

Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

COMMON MISTAKE

Do not draw a digital signal with sloping sides. A digital signal jumps *instantly*, so its edges are straight vertical lines. Also do not confuse "digital" with "electronic": an analog signal can also be electronic.

اردو میں سمجھیں

اینالاگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔ ڈیجیٹل سگنل کی صرف دو سطحیں ہوتی ہیں، HIGH (1) اور LOW (0)، اور یہ اچانک ایک سے دوسری پر جاتا ہے، اس لیے مربع لہر (Square Wave) بنتی ہے۔ ڈیجیٹل سگنل شور سے کم متاثر ہوتا ہے، اس لیے زیادہ قابلِ بھروسا ہے۔

EXAM TIP

Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave shape, reliability, example). Always label your diagrams: axes V and t, and HIGH(1)/LOW(0) on the square wave.

CHECK YOURSELF

1. What shape of wave does a digital signal make?
2. Which type of signal is more affected by noise?
3. Give one example of an analog signal.

Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An analog signal:

- a. Changes smoothly, like a sine wave
- b. Jumps only 0/1
- c. Is a truth table
- d. Is TCP

MCQ 2. A digital signal looks like:

- a. A square wave of HIGH and LOW
- b. A smooth sine
- c. A pie chart
- d. A stack

MCQ 3. Noise harms analog more because:

- a. Many in-between levels can be bent
- b. Bits cannot exist
- c. OSI has 7 layers
- d. Python is analog

MCQ 4. Human voice on a microphone is first:

- a. Analog
- b. A Boolean 1 only
- c. A K-map
- d. SQL

MCQ 5. HIGH in digital means:

- a. 1 / ON / TRUE
- b. 0 / OFF
- c. A continuous ramp
- d. Gray code

MCQ 6. Copying analog tape many times:

- a. Loses quality
- b. Stays perfect
- c. Creates bits
- d. Builds NAND

Extra short questions

1. Define analog signal and digital signal.
2. Give one example of each.
3. Why is digital more reliable?
4. What does a computer do with analog sound?

Extra long question

1. Draw analogue (sine) and digital (square) waves. Tabulate nature and reliability. This is a ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Analog: continuous, smooth change. Digital: only two levels, HIGH(1) and LOW(0), a square wave.
2. Analog: traditional thermometer / voice. Digital: computer data / logic pulse.
3. A bent 0 or 1 can be restored to a clean 0 or 1; analog distortion stays in the wave.
4. Sample and quantise it into bits, process the bits, then convert back to analog for a speaker.

Long

1. Sine labelled V against t; square labelled HIGH/LOW. Table: analog = continuous, low reliability; digital = discrete bits, high reliability. Close with the noise argument.

1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

LEARN IT

Boolean algebra was developed by the English mathematician **George Boole**. It is a kind of algebra that deals with only two values: **TRUE (1)** and **FALSE (0)**. It is the foundation of digital electronics and computer programming — every `if` condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, $Y = A + B$

Some simple rules (laws) follow directly from the meaning of AND, OR and NOT. They are useful when you check or simplify answers:

OR rules	AND rules	NOT rules
$A + 0 = A$	$A \cdot 1 = A$	$(A')' = A$
$A + 1 = 1$	$A \cdot 0 = 0$	$A + A' = 1$
$A + A = A$	$A \cdot A = A$	$A \cdot A' = 0$

WORKED EXAMPLE

Using a law: Simplify $Y = A'B + AB$.

Step 1: Both terms contain B, so take B out: $Y = B(A' + A)$.

Step 2: $A' + A = 1$ (a variable OR its opposite is always 1).

Step 3: $Y = B \cdot 1 = B$.

COMMON MISTAKE

In Boolean algebra "+" means OR, not addition. So $1 + 1 = 1$ (TRUE OR TRUE is TRUE), not 2. There is no value bigger than 1.

بولین الجبرا جارج بول (George Boole) نے بنایا۔ اس میں صرف دو قیمتیں ہوتی ہیں: درست (1) اور غلط (0)۔ متغیرات (Variables) جیسے A اور B صرف 0 یا 1 رکھتے ہیں، آپریشنز AND، OR اور NOT ہیں، اور ان کے ملاپ سے ایکسپریشن (Expression) بنتا ہے۔

CHECK YOURSELF

1. Who developed Boolean algebra?
2. In Boolean algebra, what is $1 + 1$?

Answers: 1. George Boole. 2. 1 (because + means OR).

LEARN IT

There are three basic Boolean operations. Every other operation can be made from them.

1. **AND operation ($\cdot$)** — gives output 1 **only when ALL inputs are 1**. Expression: $Y = A \cdot B$ (often written just AB).
2. **OR operation (+)** — gives output 1 **when ANY input is 1**. Expression: $Y = A + B$.
3. **NOT operation (' or $\bar{}$)** — **inverts** (flips) the input. If the input is 1 the output is 0, and if the input is 0 the output is 1. It is a **unary** operation, meaning it works on only one operand. Expression: $Y = A'$ (also written as A with a bar on top, $\bar{A}$).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

WORKED EXAMPLE

Evaluate each expression for $A = 1, B = 0$.

- (a) $A \cdot B = 1 \cdot 0 = 0$ (not all inputs are 1).
- (b) $A + B = 1 + 0 = 1$ (at least one input is 1).
- (c) $B' = 0' = 1$ (NOT flips 0 to 1).
- (d) $A \cdot B' = 1 \cdot 1 = 1$.

REMEMBER IT

AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

اردو میں سمجھیں

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ). OR آپریشن تب 1 دیتا ہے جب کوئی ایک ان پٹ بھی 1 ہو (دو دروازوں والا کمرہ). NOT آپریشن ان پٹ کو الٹ دیتا ہے اور صرف ایک ان پٹ پر کام کرتا ہے، اس لیے اسے یونری (Unary) کہتے ہیں۔

CHECK YOURSELF

1. What symbol is used for the OR operation?
2. Why is NOT called a unary operation?
3. Find $A + B'$ when $A = 0, B = 1$.

Answers: 1. "+". 2. Because it works on only one operand (one input). 3. $B' = 0$, so $0 + 0 = 0$.

LEARN IT

A **truth table** shows **all possible input combinations** and the output for each one, for a logic gate or a circuit. It is the "complete record" of how the circuit behaves.

How to create a truth table:

1. Count the number of inputs (n).
2. Calculate total rows = 2^n .
3. List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
4. Calculate the output for each combination.

2-input example ($n = 2$, rows = 4)

A	B	Combination
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3-input example ($n = 3$, rows = 8)

A	B	C	Decimal
---	---	---	---------

0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

REMEMBER IT

Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write 0,0,0,0,1,1,1,1 (change every 4 rows). For n inputs the leftmost column changes every 2^{n-1} rows.

WORKED EXAMPLE

Make the truth table of a 3-input AND: $Y = A \cdot B \cdot C$.

Step 1: $n = 3$, so rows = $2^3 = 8$.

Step 2: List the 8 combinations in binary order.

Step 3: AND gives 1 only when all inputs are 1, so only row 111 gives 1.

A	B	C	$Y = A \cdot B \cdot C$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

COMMON MISTAKE

Do not list rows in a random order — you may miss or repeat a combination. Always count in binary from 000 to 111. Also remember rows = 2^n , not $2 \times n$: 3 inputs give 8 rows, not 6.

اردو میں سمجھیں

ٹرو تھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔ اگر ان پٹ n ہوں تو قطاریں (Rows) 2 کی طاقت n ہوں گی؛ دو ان پٹ کے لیے 4 اور تین ان پٹ کے لیے 8 قطاریں۔ ان پٹ ہمیشہ بائری ترتیب میں لکھیں۔

CHECK YOURSELF

1. How many rows does a truth table with 4 inputs have?
2. In a 3-input table, what is the 6th row (counting from row 1 = 000)?

Answers: 1. $2^4 = 16$. 2. 101.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Boolean algebra uses:

- a. TRUE(1) and FALSE(0) only
- b. All real numbers
- c. ASCII letters
- d. IPv4

MCQ 2. AND is 1 only when:

- a. Every input is 1
- b. Any input is 1
- c. Inputs differ
- d. There is no input

MCQ 3. OR is 1 when:

- a. Any input is 1
- b. All inputs are 0
- c. Inputs are equal
- d. NOT is applied twice

MCQ 4. NOT of 1 is:

- a. 0
- b. 1
- c. 2
- d. A

MCQ 5. Rows in an n-input truth table:

- a. 2^n
- b. n
- c. n^2
- d. n+1

MCQ 6. Two-input truth table has:

- a. 4 rows
- b. 2 rows
- c. 8 rows
- d. 16 rows

Extra short questions

1. Who developed Boolean algebra and what two values does it use?
2. Write $Y = A \cdot B$ and $Y = A + B$ in words.
3. Why is NOT called unary?
4. List the four steps to build a truth table.

Extra long question

1. Write complete truth tables for 2-input AND, OR and 1-input NOT. Add one real-life lock/door/switch example each.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. George Boole. TRUE(1) and FALSE(0).

2. AND: output 1 only if all inputs are 1. OR: output 1 if any input is 1.
3. It has only one operand: it inverts a single input.
4. Count inputs n ; rows = 2^n ; list combinations in binary order; compute each output.

Long

1. AND 0001 on AB=11 (two locks). OR 0111 (two doors). NOT 10 / 01 (light switch). Show all four/two rows.

1.1.7 Basic Logic Gates (AND, OR, NOT) ★ Golden

LEARN IT

Logic gates are electronic circuits that perform Boolean operations. They take one or more input bits and produce one output bit. They are the **building blocks of all digital systems** — a processor contains billions of them.

The three basic gates match the three basic operations: the **AND gate**, the **OR gate** and the **NOT gate** (also called an **inverter**).



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

AND gate — $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR gate — $Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate — $Y = A'$

A	Y
0	1
1	0

WORKED EXAMPLE

A school bell circuit. The bell (Y) should ring when the timer says "period over" ($A = 1$) **and** the school is open ($B = 1$).

Step 1: The bell needs both conditions, so use an AND gate: $Y = A \cdot B$.

Step 2: Sunday, timer fires: $A = 1$, $B = 0 \rightarrow Y = 1 \cdot 0 = 0$. No bell.

Step 3: Monday, timer fires: $A = 1$, $B = 1 \rightarrow Y = 1$. The bell rings.

If instead the bell should ring when *either* the timer fires *or* the principal presses a button, you would use an OR gate.

REMEMBER IT

Shortcut: AND output is 0 if *any* input is 0. OR output is 1 if *any* input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

COMMON MISTAKE

Do not mix up the AND and OR symbols. AND has a **flat** input side (like the letter D). OR has a **curved** input side and a **pointed** output, like a shield or arrowhead.

اردو میں سمجھیں

لاجک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔ AND گیٹ صرف تب 1 دیتا ہے جب دونوں ان پٹ 1 ہوں، OR گیٹ صرف تب 0 دیتا ہے جب دونوں ان پٹ 0 ہوں، اور NOT گیٹ ان پٹ کو الٹ دیتا ہے۔

EXAM TIP

You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y. Write the full truth table in binary order. A frequent MCQ: "Which gate gives 1 only when all inputs are 1?" (AND).

CHECK YOURSELF

1. What is the output of an OR gate when $A = 0$ and $B = 0$?
2. What is another name for the NOT gate?
3. For which input does a 2-input AND gate output 1?

Answers: 1. 0. 2. Inverter. 3. Only $A = 1$, $B = 1$.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Logic gates implement:

- a. Boolean operations in hardware
- b. SQL joins
- c. HTTP only
- d. Pie charts

MCQ 2. AND output is 1 for inputs 1,1:

- a. 1
- b. 0
- c. Z
- d. X

MCQ 3. OR of 1 and 0 is:

- a. 1
- b. 0
- c. NAND
- d. XOR

MCQ 4. The inverter is the:

- a. NOT gate
- b. AND gate
- c. Bus
- d. Router

MCQ 5. Building blocks of digital systems are:

- a. Logic gates
- b. Spreadsheets
- c. MVPs
- d. Fonts

MCQ 6. $Y = A \cdot B$ is the:

- a. AND expression
- b. OR expression
- c. XOR
- d. NOR

Extra short questions

1. Define a logic gate.
2. State the AND, OR, NOT expressions.
3. Draw the truth table of 2-input AND from memory.
4. Why is NOT drawn with a bubble?

Extra long question

1. For AND, OR and NOT: symbol, expression, full truth table, and one daily-life model. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. An electronic circuit that performs a Boolean operation; the building block of digital systems.
2. $Y = A \cdot B$; $Y = A + B$; $Y = A'$ (or $\bar{A}$).
3. $00 \rightarrow 0$, $01 \rightarrow 0$, $10 \rightarrow 0$, $11 \rightarrow 1$.
4. The bubble marks inversion of the input.

Long

1. Standard IEC/US symbols. AND two locks, OR two doors, NOT a switch. Tables must have every row.

1.1.7 Universal & Advanced Logic Gates

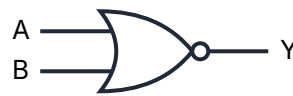
LEARN IT

NAND means "NOT AND": it is an AND gate followed by a NOT. **NOR** means "NOT OR": an OR gate followed by a NOT. On the symbol, the NOT is shown as a small bubble at the output.

NAND and NOR are called **universal gates** because **any logic circuit in the world can be built using only NAND gates (or only NOR gates)**. Factories like this: they can mass-produce one type of chip and build everything from it.



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

NAND (NOT AND) — $Y = (A \cdot B)'$

A	B	$A \cdot B$	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOR (NOT OR) — $Y = (A + B)'$

A	B	$A + B$	Y
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

WORKED EXAMPLE

Show that a NAND gate can work as a NOT gate.

Step 1: Connect the same input A to both inputs of the NAND gate. Then $Y = (A \cdot A)'$.

Step 2: Since $A \cdot A = A$, we get $Y = A'$.

Step 3: Check: $A = 0 \rightarrow (0 \cdot 0)' = 0' = 1$. $A = 1 \rightarrow (1 \cdot 1)' = 1' = 0$. This is exactly the NOT truth table.

In the same way, a NOR gate with both inputs tied together gives $(A + A)' = A'$. And a NAND followed by a NAND-as-NOT gives $((A \cdot B)')' = A \cdot B$, an AND gate. This is why they are "universal".

REMEMBER IT

NAND is the exact **opposite** of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

COMMON MISTAKE

Do not forget the bubble. Without the small circle, a NAND symbol becomes an AND symbol and your answer is wrong. Also, $(A \cdot B)'$ is **not** the same as $A' \cdot B'$: for $A = 1, B = 0$, $(1 \cdot 0)' = 1$ but $0 \cdot 1 = 0$.

اردو میں سمجھیں

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR۔ انہیں یونیورسل (عالمگیر) گیٹ کہتے ہیں کیونکہ صرف NAND یا صرف NOR گیٹس سے دنیا کا کوئی بھی لاجک سرکٹ بنایا جا سکتا ہے۔
NAND صرف تب 0 دیتا ہے جب دونوں ان پٹ 1 ہوں، اور NOR صرف تب 1 دیتا ہے جب دونوں ان پٹ 0 ہوں۔

CHECK YOURSELF

1. Why are NAND and NOR called universal gates?
2. What is the NOR output for $A = 0, B = 1$?
3. How do you make a NOT gate from a NAND gate?

Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.

LEARN IT

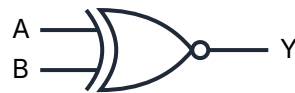
XOR (Exclusive OR) gives output 1 when the inputs are **different**. Expression: $Y = A \oplus B$. In terms of basic operations, $A \oplus B = A'B + AB'$.

XNOR (Exclusive NOR) is XOR followed by NOT. It gives output 1 when the inputs are the **same**. Expression: $Y = (A \oplus B)' = A'B' + AB$.

The XOR symbol is an OR symbol with an extra curved line at the input side. XNOR adds a bubble at the output.



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

$$\text{XOR} — Y = A \oplus B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$\text{XNOR} — Y = (A \oplus B)'$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

All seven gates at a glance

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

(NOT has one input: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.)

WORKED EXAMPLE

A staircase light. A light has one switch at the bottom (A) and one at the top (B) of a staircase. Flipping either switch should change the light.

Step 1: Start with both switches down: $A = 0$, $B = 0$, light off ($Y = 0$).

Step 2: Flip the bottom switch: $A = 1$, $B = 0 \rightarrow$ light on ($Y = 1$).

Step 3: Walk up and flip the top switch: $A = 1$, $B = 1 \rightarrow$ light off ($Y = 0$).

Step 4: The pattern 0, 1, 1, 0 is the XOR truth table. So the light is $Y = A \oplus B$.

REMEMBER IT

Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the **same**. XNOR is sometimes called the "equality gate".

COMMON MISTAKE

XOR is not the same as OR. They differ only in the last row: for $A = 1, B = 1$, OR gives 1 but XOR gives 0.

اردو میں سمجھیں

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔ XNOR دراصل XOR کے بعد NOT ہے۔ OR اور XOR میں فرق صرف آخری قطار میں ہے جہاں دونوں ان پٹ 1 ہوں۔

CHECK YOURSELF

1. What is $1 \oplus 1$?
2. Which gate outputs 1 when both inputs are the same?
3. Write XOR using AND, OR and NOT.

Answers: 1. 0. 2. XNOR. 3. $A \oplus B = A'B + AB'$.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. NAND and NOR are universal because:

- a. Any circuit can be built from only that gate
- b. They are analog
- c. They store files
- d. They sort arrays

MCQ 2. NAND is:

- a. NOT of AND
- b. AND of NOT
- c. OR of XOR
- d. Only OR

MCQ 3. NAND of 1,1 is:

- a. 0
- b. 1
- c. Z
- d. 2

MCQ 4. NOR of 0,0 is:

- a. 1
- b. 0
- c. A
- d. B

MCQ 5. XOR is 1 when inputs are:

- a. Different
- b. Both 1 only
- c. Both 0 only
- d. Always 1

MCQ 6. XNOR is 1 when inputs are:

- a. The same
- b. Different
- c. Only 1,0
- d. Floating

Extra short questions

1. Why are NAND and NOR called universal gates?
2. Write NAND and NOR expressions.
3. Memory trick for XOR vs XNOR.
4. Truth table of XOR.

Extra long question

1. Show how a NOT, AND and OR can each be built from NAND only. Include truth-table checks.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Any Boolean function (AND, OR, NOT, XOR...) can be built using only NAND, or only NOR.
2. $Y = (A \cdot B)'$ and $Y = (A + B)'$.
3. XOR: eXclusively different $\rightarrow 1$. XNOR: same $\rightarrow 1$ (equivalence).
4. $00 \rightarrow 0$, $01 \rightarrow 1$, $10 \rightarrow 1$, $11 \rightarrow 0$.

Long

1. NOT: NAND with inputs tied. AND: NAND then NAND-NOT. OR: invert inputs with NAND-NOTs then NAND (De Morgan). Tables must match the target gate.

1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams ★ Golden

LEARN IT

A **logic expression** (Boolean expression) combines variables with operations, for example $Y = A + B' \cdot C$. To evaluate it correctly you must follow the **order of precedence**, just like BODMAS in maths:

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

WORKED EXAMPLE

Worked example 1: truth table of $Y = A + B' \cdot C$

Step 1: 3 inputs, so 8 rows.

Step 2: Precedence: first find B' (NOT), then $B' \cdot C$ (AND), then $A + (B' \cdot C)$ (OR). Make a column for each step.

A	B	C	B'	$B' \cdot C$	$Y = A + B' \cdot C$
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

Step 3: $Y = 1$ in rows 1, 4, 5, 6, 7 (decimal). Check one row by hand: $A=0, B=0, C=1 \rightarrow B' = 1, B' \cdot C = 1, Y = 0 + 1 = 1$. ✓

WORKED EXAMPLE

Worked example 2: truth table of $Y = (A + B) \cdot C'$

Step 1: Parentheses first: find $A + B$. Then NOT: C' . Then AND the two.

A	B	C	A+B	C'	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

Step 2: $Y = 1$ in rows 2, 4 and 6 (decimal).

Minterms (SOP) and Maxterms (POS)

LEARN IT

Sometimes you are given a truth table and must *find* the expression. There are two standard ways.

Minterm (used in SOP — Sum of Products): a **product (AND)** of all the variables, written for a row whose output is **1**. A variable is **complemented (A')** if its value is 0, and **uncomplemented (A)** if its value is 1. OR-ing all the minterms gives the SOP expression.

Maxterm (used in POS — Product of Sums): a **sum (OR)** of all the variables, written for a row whose output is **0**. A variable is **complemented (A')** if its value is 1, and **uncomplemented (A)** if its value is 0. AND-ing all the maxterms gives the POS expression.

A	B	Output (Y)	Minterm (SOP)	Maxterm (POS)
0	0	0	–	$A + B$
0	1	1	$A'B$	–
1	0	0	–	$A' + B$
1	1	1	AB	–

Final SOP expression: $Y = A'B + AB$. **Final POS expression:** $Y = (A + B)(A' + B)$. Both describe the same table. (Both simplify to $Y = B$ — look at the table: Y always equals B. You will confirm this with a K-map later.)

WORKED EXAMPLE

Worked example 3: write the SOP for a 3-input "majority" circuit. $Y = 1$ when at least two of A, B, C are 1.

Row	A	B	C	Y	Minterm
0	0	0	0	0	–
1	0	0	1	0	–
2	0	1	0	0	–
3	0	1	1	1	$A'BC$
4	1	0	0	0	–
5	1	0	1	1	$AB'C$
6	1	1	0	1	ABC'
7	1	1	1	1	ABC

Step 1: Pick the rows with $Y = 1$: rows 3, 5, 6, 7.

Step 2: Row 3 is $A=0, B=1, C=1 \rightarrow A$ is 0 so write A' ; B and C are 1 so write B, C $\rightarrow A'BC$.

Step 3: Do the same for the other rows and OR them:

$Y = A'BC + AB'C + ABC' + ABC$. In short form: $Y = \Sigma m(3, 5, 6, 7)$. We will simplify this with a K-map in 1.1.13.

$$Y = A'BC + AB'C + ABC' + ABC = \Sigma m(3, 5, 6, 7)$$

REMEMBER IT

SOP = 1s, POS = 0s. Minterm: 0 $\rightarrow$ bar, 1 $\rightarrow$ plain. Maxterm: the reverse (1 $\rightarrow$ bar, 0 $\rightarrow$ plain). Think "Min likes 1s, Max likes 0s".

COMMON MISTAKE

Two classic errors: (1) writing minterms for the 0-rows — minterms come only from rows where $Y = 1$; (2) forgetting precedence — in $A + B \cdot C$ you must do $B \cdot C$ first. $A + B \cdot C$ is **not** $(A + B) \cdot C$: for $A=1, B=0, C=0$ the first gives 1 and the second gives 0.

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR۔ منٹرم (Minterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 1 ہو؛ اس میں 0 والے متغیر پر بار (A) لگتی ہے۔ میکس ٹرم (Maxterm) اُن قطاروں سے بنتا ہے جن کا آؤٹ پٹ 0 ہو؛ اس میں 1 والے متغیر پر بار لگتی ہے۔

CHECK YOURSELF

1. Which comes first: AND or OR?
2. Write the minterm for $A = 1, B = 0, C = 1$.
3. Write the maxterm for $A = 0, B = 1$.

Answers: 1. AND. 2. $AB'C$. 3. $A + B'$.

LEARN IT

A **logic diagram** shows how logic gates are connected to create a circuit. It is a "picture" of a Boolean expression. To draw one, use the order of precedence: the operation with the **highest priority is drawn first** (nearest the inputs), and the last operation is the gate that gives the final output.

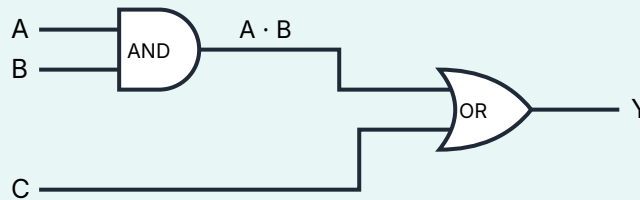
WORKED EXAMPLE

Example 1: Draw the logic diagram for $Y = A \cdot B + C$.

Step 1: Find the operations: AND ($\cdot$) and OR ($+$).

Step 2: AND has higher precedence than OR, so draw the AND gate first, with inputs A and B. Its output is $A \cdot B$.

Step 3: Draw an OR gate. Feed it $A \cdot B$ and C. Its output is Y.



$Y = A \cdot B + C$: AND first, then OR.

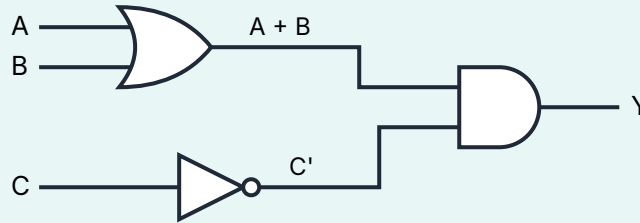
WORKED EXAMPLE

Example 2: Draw the logic diagram for $Y = (A + B) \cdot C'$.

Step 1: Parentheses have the highest priority, so draw the OR gate for $A + B$ first.

Step 2: NOT is next: draw a NOT gate on input C to get C' .

Step 3: The last operation is AND: feed $(A + B)$ and C' into an AND gate to get Y .



$Y = (A + B) \cdot C'$: OR and NOT first, then AND.

You can check this circuit against the truth table in 1.1.8 (worked example 2): $Y = 1$ only for rows 010, 100 and 110.

REMEMBER IT

Read the expression and ask "What is done **last**?" That gate goes on the far right, next to Y . Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2 gates; $Y = (A+B) \cdot C'$ needs 3.

COMMON MISTAKE

Do not draw a separate wire for a variable each time it appears and then leave one unconnected. Every gate input must be joined to an input variable or to another gate's output, and every wire should be clearly labelled in exams.

اردو میں سمجھیں

لاجک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔ ترجیحی ترتیب کے مطابق جو آپریشن پہلے ہوتا ہے اس کا گیٹ پہلے (ان پٹ کے قریب) بنایا جاتا ہے، اور آخری آپریشن کا گیٹ آؤٹ پٹ Y دیتا ہے۔ مثلاً $Y = A \cdot B + C$ میں پہلے AND گیٹ اور پھر OR گیٹ بنتا ہے۔

EXAM TIP

Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols with labelled intermediate outputs (like $A \cdot B$), and finish with the truth table. Marks are lost for wrong gate shapes and missing bubbles.

CHECK YOURSELF

1. In $Y = A + B \cdot C$, which gate is drawn first?
2. How many gates are needed for $Y = A' \cdot B$?

Answers: 1. The AND gate (for $B \cdot C$). 2. Two: one NOT gate (for A') and one AND gate.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Precedence (highest first) is:

- a. () then NOT then AND then OR
- b. OR then AND then NOT
- c. AND then () then OR
- d. XOR first always

MCQ 2. A minterm is used when Y is:

- a. 1 (SOP)
- b. 0 (POS)
- c. Always 0
- d. A stack

MCQ 3. For $AB=01$ if $Y=1$ the minterm is:

- a. $A'B$
- b. AB
- c. $A+B$
- d. $A'B'$

MCQ 4. A maxterm is a:

- a. Sum (OR) for a 0 output
- b. Product for a 1
- c. K-map only
- d. Gate delay

MCQ 5. SOP means:

- a. Sum of products
- b. Sum of people
- c. Stack of pointers
- d. Set of pies

MCQ 6. In a logic diagram you draw first the operation with:

- a. Highest precedence
- b. Lowest precedence
- c. Most inputs
- d. Green ink

Extra short questions

1. State operator precedence in Boolean algebra.
2. Define minterm and maxterm.
3. From Y rows 01 and 11 = 1, write SOP.
4. For $Y = A \cdot B + C$, which gate is drawn first?

Extra long question

1. Given a 2-input truth table with $Y = 0,1,0,1$, write minterms, SOP, maxterms that apply, and draw the logic diagram.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Parentheses, NOT, AND, OR.
2. Minterm: AND of all variables (complemented if 0) for rows where $Y=1$. Maxterm: OR of all variables (complemented if 1) for rows where $Y=0$.
3. $Y = A'B + AB$.
4. AND ($A \cdot B$), then OR with C, because AND beats OR.

Long

1. $Y=1$ on 01 and 11 $\rightarrow A'B + AB$ which simplifies to B. Diagram: if unsimplified, two ANDs into an OR; simplified, $Y=B$ (wire). Show both and say examiners accept the simplified form if working is shown.

1.1.13 Karnaugh Maps (K-Maps) Simplification ★ Golden

LEARN IT

A **Karnaugh map (K-map)** is a graphical tool used to **simplify Boolean expressions**. It is a grid of cells — one cell for each row of the truth table. The cells are arranged so that **neighbouring (adjacent) cells differ by only ONE bit**. This ordering is called **Gray code**.

You write the output (1 or 0) of each row into its cell, then **group the 1s** in rectangles of size 1, 2, 4 or 8 (powers of 2). Each group becomes one simple term. This simplifies the equation without long algebra.

Two-variable K-map (A, B)

2 variables need $2^2 = 4$ cells.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Three-variable K-map (A, B, C)

3 variables need $2^3 = 8$ cells. The columns follow the Gray code sequence **00, 01, 11, 10** (not 00, 01, 10, 11). This makes adjacent cells differ by only one variable.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	m_0 (A'B'C')	m_1 (A'B'C)	m_3 (A'BC)	m_2 (A'BC')
A = 1	m_4 (AB'C')	m_5 (AB'C)	m_7 (ABC)	m_6 (ABC')

LEARN IT

Grouping rules

1. Group only 1s. Groups must be rectangles of size 1, 2, 4 or 8 — never 3, 5 or 6.
2. Make groups **as large as possible**. Bigger groups give simpler terms.
3. Use **as few groups as possible**, but every 1 must be in at least one group.
4. **Overlapping is allowed** — a 1 may be used in more than one group.
5. **Wrapping around the edges is allowed**: a cell in the far-left column is adjacent to the cell in the far-right column (BC = 00 is next to BC = 10).
6. To read a group: variables that **change** (0 to 1) inside the group are **eliminated**. Variables that stay the same are kept — plain if they stay 1, complemented if they stay 0.

Group size tells you how many variables disappear: in a 3-variable map, a group of 2 leaves 2 variables, a group of 4 leaves 1 variable, and a group of 8 means $Y = 1$.

WORKED EXAMPLE

K-map example 1 (2 variables): simplify $Y = A'B + AB$.

Step 1: The minterms are $A'B$ (cell 1) and AB (cell 3). Put 1 in those cells, 0 elsewhere.

	B = 0	B = 1
A = 0	0	1
A = 1	0	1

Step 2: The two 1s are in the same column, one above the other. Make one group of 2.

Step 3: Inside the group, A changes (0 then 1) → eliminate A. B stays 1 → keep B.

Result: $Y = B$. This agrees with the algebra in 1.1.4 and the minterm table in 1.1.8.

WORKED EXAMPLE

K-map example 2 (2 variables, overlapping): simplify $Y = A'B' + A'B + AB$.

Step 1: Put 1s in cells 0, 1 and 3.

	B = 0	B = 1
A = 0	1	1
A = 1	0	1

Step 2: Group 1 = the top row (cells 0 and 1). A stays 0, B changes → term **A'** .

Step 3: Group 2 = the right column (cells 1 and 3). A changes, B stays 1 → term **B** . Cell 1 is used twice — overlapping is allowed.

Result: $Y = A' + B$. Check: $A' + B$ is 0 only when $A = 1$ and $B = 0$, which is exactly the one 0 cell (cell 2). ✓

WORKED EXAMPLE

K-map example 3 (3 variables): simplify the majority function $Y = \sum m(3, 5, 6, 7) = A'BC + AB'C + ABC' + ABC$.

Step 1: Fill the map. Careful — the columns are in Gray order 00, 01, 11, 10, so m_3 is in the third column and m_2 in the fourth.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	0	1 (m_3)	0
A = 1	0	1 (m_5)	1 (m_7)	1 (m_6)

Step 2: No group of 4 is possible (there is no 2×2 block or full row of 1s). So make groups of 2, each using m_7 :

- Group 1: $m_3 + m_7$ (column BC = 11). A changes; B = 1, C = 1 stay $\rightarrow$ **BC**.
- Group 2: $m_5 + m_7$ (row A = 1, columns 01 and 11). B changes; A = 1, C = 1 stay $\rightarrow$ **AC**.
- Group 3: $m_7 + m_6$ (row A = 1, columns 11 and 10). C changes; A = 1, B = 1 stay $\rightarrow$ **AB**.

Result: $Y = AB + BC + AC$. Four 3-variable terms (needing 4 three-input AND gates) became three 2-variable terms.

Check one row: A=1, B=0, C=1 $\rightarrow$ AB = 0, BC = 0, AC = 1 $\rightarrow$ Y = 1. In the truth table, row 5 is 1. ✓

WORKED EXAMPLE

K-map example 4 (3 variables, wrap-around): simplify $Y = \sum m(0, 2, 4, 6)$.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	1	0	0	1
A = 1	1	0	0	1

Step 1: The 1s are in the far-left column (BC = 00) and the far-right column (BC = 10).

Step 2: Because the map wraps around, these two columns are adjacent. Make **one group of 4**: m_0, m_4, m_2, m_6 .

Step 3: In this group A changes (0, 1) and B changes (0, 1). Only C stays the same: C = 0 $\rightarrow$ term **C'**.

Result: $Y = C'$. Without wrapping you would get two groups ($B'C' + BC'$), which is correct but not the simplest.

WORKED EXAMPLE

K-map example 5 (3 variables): simplify $Y = \sum m(1, 4, 5, 6, 7)$ — this is the output of $Y = A + B'C$ from 1.1.8.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	0	0
A = 1	1	1	1	1

Step 1: Group of 4: the whole row $A = 1$ (m_4, m_5, m_7, m_6). B and C both change; A stays 1 $\rightarrow$ **A**.

Step 2: m_1 is still alone. Pair it with m_5 below it (column BC = 01). A changes; B = 0, C = 1 stay $\rightarrow$ **B'C**.

Result: $Y = A + B'C$ — the K-map confirms the original expression was already in its simplest form.

WORKED EXAMPLE

Going further — a 4-variable K-map: simplify $Y = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$. (Not required by the slides, but a good test of the same rules.) 4 variables need 16 cells; both rows (AB) and columns (CD) use Gray order 00, 01, 11, 10.

	CD = 00	CD = 01	CD = 11	CD = 10
AB = 00	1 (m_0)	0	0	1 (m_2)
AB = 01	0	1 (m_5)	1 (m_7)	0
AB = 11	0	1 (m_{13})	1 (m_{15})	0
AB = 10	1 (m_8)	0	0	1 (m_{10})

Step 1: The centre 2×2 block (m_5, m_7, m_{13}, m_{15}) is a group of 4. A and C change; B = 1 and D = 1 stay $\rightarrow$ **BD**.

Step 2: The four corners (m_0, m_2, m_8, m_{10}) are adjacent through wrap-around (left–right and top–bottom). A and C change; B = 0 and D = 0 stay $\rightarrow$ **B'D'**.

Result: $Y = BD + B'D'$ (this is B XNOR D). Eight 4-variable minterms became two 2-variable terms.

REMEMBER IT

Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step. Imagine the K-map is drawn on a tube, so the left and right edges touch. For reading groups: "If it changes, it goes."

COMMON MISTAKE

Most common K-map mistakes: (1) writing the columns as 00, 01, 10, 11 — this puts m_2 and m_3 in the wrong places; (2) making groups of 3 or 6 — only 1, 2, 4, 8 are allowed; (3) grouping diagonally — groups must be rectangles of side-by-side cells; (4) stopping at small groups when a bigger group (or a wrap-around group) is possible.

اردو میں سمجھیں

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔ اس کے ساتھ والے خانوں میں صرف ایک ہٹ کا فرق ہوتا ہے (گرے کوڈ: 00, 01, 10, 11)۔ 1 والے خانوں کو 1, 2, 4 یا 8 کے گروپوں میں جتنا بڑا ممکن ہو جوڑیں؛ گروپ ایک دوسرے پر چڑھ سکتے ہیں اور کناروں سے گھوم کر بھی بن سکتے ہیں۔ گروپ کے اندر جو متغیر بدلتا ہے وہ ختم ہو جاتا ہے۔

EXAM TIP

K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s too), circle the groups clearly, write the term for each group beside it, and give the final simplified expression. Examiners often add "draw the logic diagram of the result" — so practise drawing it too.

CHECK YOURSELF

1. In a 3-variable K-map, which column is next to $BC = 01$ on the right?
2. Simplify $Y = \sum m(2, 3)$ for two variables A, B.
3. Simplify $Y = \sum m(4, 5, 6, 7)$ for three variables.

Answers: 1. $BC = 11$. 2. Cells 2 and 3 are the row $A = 1$; B changes $\rightarrow Y = A$. 3. The whole row $A = 1$ is a group of 4 $\rightarrow Y = A$.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Adjacent K-map cells differ by:

- a. One bit (Gray code)
- b. Two bits
- c. The output only
- d. ASCII

MCQ 2. Group sizes must be:

- a. Powers of 2 (1,2,4,8...)
- b. Only 3
- c. Only 5
- d. Prime numbers

MCQ 3. A 3-variable K-map has:

- a. 8 cells
- b. 4 cells
- c. 3 cells
- d. 16 cells

MCQ 4. Wrapping left to right is:

- a. Allowed
- b. Forbidden
- c. Only for XOR
- d. Only analog

MCQ 5. Inside a group, variables that change are:

- a. Eliminated
- b. Doubled
- c. ANDed twice
- d. Sent to OR first

MCQ 6. We group:

- a. 1s (for SOP)
- b. Only 0s always
- c. Gray cells empty
- d. Page numbers

Extra short questions

1. What is a K-map?
2. State the grouping rules.
3. Gray code order for two bits.
4. Why not group three 1s as a triple?

Extra long question

1. Simplify $Y(A,B,C) = \sum m(0, 2, 4, 6)$ with a 3-variable K-map. Show cells, groups and the final expression.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A grid for simplifying Boolean expressions. Adjacent cells differ by one bit (Gray code).
2. Groups of 1,2,4,8...; as large as possible; overlap allowed; wrap around edges; drop variables that change inside the group.
3. 00, 01, 11, 10.
4. 3 is not a power of 2, so it is not a valid grouping.

Long

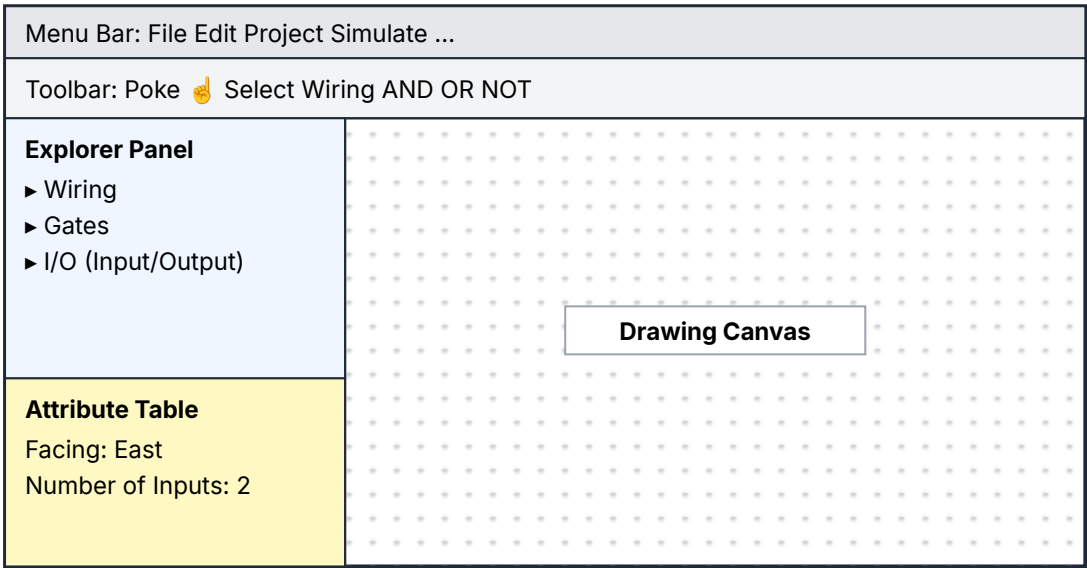
1. Cells m_0, m_2, m_4, m_6 are 1 (even minterms). They form a quad: C is 0 in all, A and B change $\rightarrow Y = C'$. Label Gray columns 00 01 11 10. Draw the wrap/quad clearly.

LogiSim Simulation Tool Guide

LEARN IT

LogiSim Evolution is a free graphical tool for **designing and simulating logic circuits**. "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change. It lets you see the truth tables and Boolean expressions you learn in theory.

1. The interface areas



The main areas of the LogiSim Evolution window.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.

Wiring colours during simulation

Colour	Meaning
--------	---------

Bright green	Logic 1 (HIGH)
Dark green	Logic 0 (LOW)
Blue	Unknown / unconnected pin
Red	Error / short circuit

REMEMBER IT

Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is wrong".

COMMON MISTAKE

If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin. Zoom in and make sure every wire end sits exactly on a connection point.

اردو میں سمجھیں

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔ اس میں مینو بار، ٹول بار، ایکسپلورر پینل، ایٹری بیوٹ ٹیبل اور ڈرائنگ کینوس ہوتے ہیں۔ چمکدار سبز تار 1، گہری سبز 0، نیلی تار غیر منسلک اور سرخ تار غلطی کو ظاہر کرتی ہے۔

CHECK YOURSELF

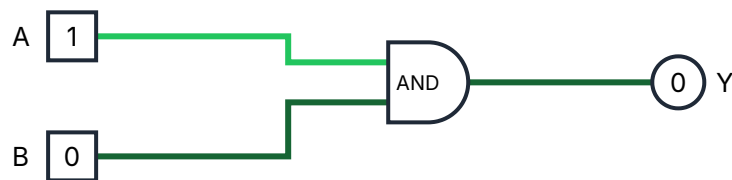
1. Where do you change the number of inputs of a gate?
2. What does a red wire mean?

Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.

WORKED EXAMPLE

Step by step: building $Y = A \cdot B$

1. Select the **AND gate** from the Toolbar (or from the Gates library) and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate. Label them A and B.
3. Select the **Output Pin** (circle icon with a 0) and place it to the right. Label it Y.
4. Use the **Wiring Tool** to click and drag wires from the input pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the output pin Y.
6. Select the **Poke Tool** (pointing finger) and click inputs A and B to toggle them between 0 and 1. Test the circuit!



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

WORKED EXAMPLE

Testing the circuit = making its truth table. Poke the inputs through all four combinations and write what Y shows:

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green

This matches the AND truth table, so the circuit is correct. Try the same with $Y = A \cdot B + C$ from 1.1.11 (add a second gate and a third input pin).

REMEMBER IT

Input pin = **square** (you push it). Output pin = **circle** (it shows a light). Poke = finger — you "poke" inputs to change them.

COMMON MISTAKE

Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it. Switch to the Poke Tool first.

اردو میں سمجھیں

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائرنگ ٹول سے تاریں جوڑیں۔ آخر میں پوک ٹول (انگلی) سے ان پٹ کو 0 اور 1 کر کے سرکٹ کو جانچیں۔

CHECK YOURSELF

1. Which icon is the input pin: square or circle?
2. Which tool do you use to change an input between 0 and 1?

Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. LogiSim Evolution is used to:

- a. Draw and simulate logic circuits
- b. Edit videos
- c. Send email
- d. Compile Java only

MCQ 2. The canvas is where you:

- a. Place and wire gates
- b. Install Python
- c. Write SQL
- d. Draw pie charts

MCQ 3. Poke tool is for:

- a. Toggling inputs while simulating
- b. Deleting the OS
- c. Printing Urdu
- d. Sorting

MCQ 4. Attribute table sets:

- a. Inputs, facing, labels
- b. Big O
- c. IP addresses
- d. Passwords

MCQ 5. To build $Y=A \cdot B$ you need:

- a. AND gate plus two inputs and an output
- b. Only XOR
- c. A stack
- d. Pandas

MCQ 6. Simulation lets you check:

- a. The truth table live
- b. HTTP status
- c. File modes
- d. HCI wireframes only

Extra short questions

1. Name four areas of the LogiSim window.
2. Steps to build $Y=A \cdot B$.
3. Where do you change the number of inputs on a gate?
4. Why simulate at all if the truth table is already on paper?

Extra long question

1. Describe, with a canvas sketch, how you would build and test $Y = A \cdot B + C$ in LogiSim.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Menu bar, toolbar, explorer (libraries), attribute table, drawing canvas.
2. Place AND; add two input pins; add output pin; wire; label; simulate with Poke.

3. Attribute table (bottom left).

4. Catches wiring mistakes and shows students the table “come alive”.

Long

1. AND for A,B; OR with C; three inputs, one output; wiring; poke all 8 combinations; tick the truth table. Mention facing and labels.

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

LEARN IT

The **Software Development Life Cycle (SDLC)** is a structured process that guides software development from beginning to end. It ensures **clear goals, better planning, risk reduction and high quality**. Without it, teams forget features, go over budget and deliver buggy software.

SDLC has **6 phases**:



Phases 1–3 in detail

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Phases 4–6 in detail

Phase	Activities and details	Deliverable
4. Testing	Find and fix bugs (errors), using Black Box and White Box testing.	Test Reports, bug-free software
5. Deployment	Release the software to end-users: Direct, Phased, or Pilot release.	Live Software
6. Maintenance	Provide support and updates, and fix post-release issues.	Patches, Updated Versions

Deployment methods: In a **direct** release the old system is switched off and the new one switched on at once. In a **phased** release the new system is introduced part by part. In a **pilot** release one small group (for example one branch or one class) uses it first before everyone else.

Testing methods

Black Box Testing

- The tester does **not** see the code.
- Tests functionality: give inputs, check outputs.
- Done by dedicated testers.

White Box Testing

- The tester **can** see the code.
- Tests the internal logic and structure.
- Done by developers.

WORKED EXAMPLE

Building a house = SDLC.

Phase 1 (Requirements): you ask the family what rooms they need — 3 bedrooms, a kitchen, a garage.

Phase 2 (Design): the architect draws the blueprints.

Phase 3 (Implementation): the builders lay the bricks.

Phase 4 (Testing): an inspector checks the wiring and the plumbing.

Phase 5 (Deployment): the family moves in.

Phase 6 (Maintenance): repairs, painting and a new room added later.

WORKED EXAMPLE

Black box vs white box on a login screen. A black-box tester types a correct password and a wrong password and checks that the first is accepted and the second rejected — without looking at the code. A white-box tester (the developer) reads the code and checks that every `if` branch, for example the "account locked after 3 tries" branch, actually runs correctly.

REMEMBER IT

Remember the order with "**R**ead **D**evelopers **I**mplement, **T**est, **D**eploy, **M**aintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass box (you can see inside).

COMMON MISTAKE

Do not say the SDLC ends at deployment. Maintenance is a full phase and often lasts the longest. Also, the SRS is the deliverable of Requirement Analysis, not of Design.

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ٹیسٹنگ، تنصیب (Deployment) اور دیکھ بھال (Maintenance)۔ پہلا مرحلہ سب سے اہم ہے اور اس سے SRS دستاویز بنتی ہے۔ بلیک باکس ٹیسٹنگ میں کوڈ نظر نہیں آتا جبکہ وائٹ باکس ٹیسٹنگ میں کوڈ دیکھ کر اندرونی منطق جانچی جاتی ہے۔

EXAM TIP

"Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of each (SRS, Design Document, Source Code, Test Reports, Live Software, Patches). A short question often asks you to differentiate black box and white box testing — answer in a 3-row table (see code?, what is tested?, who does it?).

CHECK YOURSELF

1. Which phase produces the SRS document?
2. Who usually performs white box testing?
3. Name the three ways to deploy software.

Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. SDLC is:

- a. A structured process from idea to retired software
- b. A logic gate
- c. A search algorithm
- d. A chart type

MCQ 2. The first SDLC phase is usually:

- a. Requirement analysis
- b. Maintenance
- c. Coding only
- d. Retirement

MCQ 3. A deliverable is:

- a. The document or product of a phase
- b. A Boolean 1
- c. A pie slice
- d. An IP port

MCQ 4. Testing comes:

- a. After implementation, before deployment
- b. Before requirements
- c. Instead of design
- d. Never

MCQ 5. Maintenance is:

- a. Fixing and updating after release
- b. The first interview
- c. NAND
- d. Big O

MCQ 6. SDLC exists to:

- a. Control quality, cost and time
- b. Replace Boolean algebra
- c. Draw K-maps
- d. Sort arrays

Extra short questions

1. Define SDLC.
2. List the six phases in order.
3. Name one deliverable of requirements and of design.
4. Why not jump straight to coding?

Extra long question

1. Tabulate the six SDLC phases with activities and one deliverable each. ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Software Development Life Cycle: a structured process that guides software from beginning to end.
2. Requirements, design, implementation/coding, testing, deployment, maintenance.
3. SRS / requirements document; design diagrams or architecture document.
4. Wrong requirements become expensive bugs; SDLC catches them on paper first.

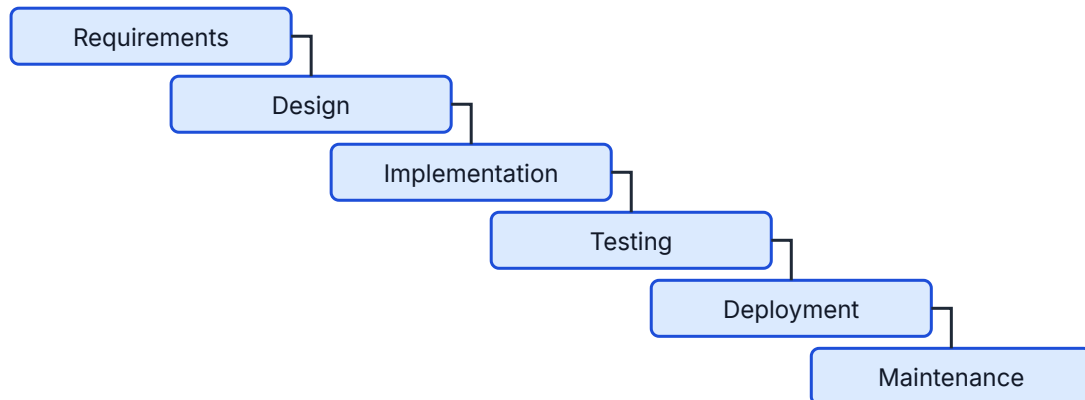
Long

1. Phase | activities | deliverable. Stress that testing uses the requirements as the oracle, and maintenance is the longest phase in real life.

1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies ★ Golden

LEARN IT

The **Waterfall model** is a **linear, sequential** approach to the SDLC. Each phase must be completed before the next one begins, like water falling down steps. There is no easy way to go back to a previous phase.



The Waterfall model: each phase "falls" into the next, in one direction only.

When to use it

- Requirements are clear, fixed and well understood.
- The project is small to medium in size.
- The technology is stable (not new or experimental).

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.

WORKED EXAMPLE

Case study: Al-Noor Library Management System. The textbook uses this system as a Waterfall case study because the requirements were **100% fixed before coding began**.

Step 1: The librarian listed exactly what was needed: add books, issue books, return books, calculate fines. Nothing would change.

Step 2: Because the list was fixed, the team could finish the full design, then all the coding, then all the testing, in order.

Step 3: Every phase produced a document, so the library has a complete record for future maintenance.

REMEMBER IT

Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

COMMON MISTAKE

Do not choose Waterfall for a project where the customer is still unsure what they want. A late change means going back through design, coding and testing, which is slow and costly.

اردو میں سمجھیں

واٹر فال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔ یہ تب بہتر ہے جب ضروریات واضح اور طے شدہ ہوں، جیسے انور لائبریری مینجمنٹ سسٹم میں۔ یہ سمجھنے میں آسان ہے مگر تبدیلیوں کے لیے لچکدار نہیں اور ٹیسٹنگ دیر سے ہوتی ہے۔

EXAM TIP

Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to use" conditions, and give at least two pros and two cons. Mention the Al-Noor Library case study as your example.

CHECK YOURSELF

1. Why is the Waterfall model called "sequential"?
2. Why did the Al-Noor Library system use Waterfall?

Answers: 1. Because each phase must be completed before the next begins, in a fixed order.
2. Because its requirements were 100% fixed before coding began.

LEARN IT

The **Agile model** is an **iterative** approach: the software is built in small, repeated cycles called **sprints** (usually 2–4 weeks long). Each sprint plans, builds and tests a small part of the software, and then shows it to the customer. **Customer feedback is gathered continuously** and used in the next sprint.



When to use it

- Requirements are unclear or may change.
- The customer wants frequent progress updates (early delivery).
- The project is complex or innovative.

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

WORKED EXAMPLE

Case study: Al-Noor Student Mobile Application. This app used Agile because teachers, parents and students **constantly requested new features** during development.

Sprint 1 (2 weeks): build login and the timetable screen. Show it to users.

Feedback: parents ask, "Can we see attendance too?"

Sprint 2: add attendance. Teachers now ask for homework notifications.

Sprint 3: add notifications. Each sprint delivers a working app, and new ideas are welcomed instead of causing a restart.

REMEMBER IT

Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

COMMON MISTAKE

"Less documentation" does not mean "no documentation" or "no planning". Agile teams still plan every sprint; they simply value working software over long documents.

اردو میں سمجھیں

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرنٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔ ہر اسپرنٹ کے بعد گاہک کی رائے لی جاتی ہے۔ یہ تبدیلیوں کے لیے بہت لچکدار ہے اور جلد کام کرنے والا سافٹ ویئر دیتا ہے، مگر اس میں دستاویزات کم ہوتی ہیں اور وقت و لاگت کا اندازہ مشکل ہوتا ہے۔ النور اسٹوڈنٹ موبائل ایپ اس کی مثال ہے۔

EXAM TIP

Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous customer feedback". Use the Al-Noor Student Mobile App as your case study.

CHECK YOURSELF

1. What is a sprint and how long does it usually last?
2. Give one disadvantage of Agile.

Answers: 1. A short development cycle that delivers a working part of the software; usually 2–4 weeks. 2. Less documentation, or unpredictable end-time and cost.

LEARN IT

Both models follow the same SDLC phases. The difference is **how** they move through them: Waterfall goes through each phase **once**, in a straight line; Agile goes through all phases **many times**, in small loops.

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

WORKED EXAMPLE

Choosing a model — think like the textbook.

Case A: Al-Noor Library Management System. Requirements fixed 100% before coding → no changes expected → **Waterfall**.

Case B: Al-Noor Student Mobile Application. Teachers, parents and students keep asking for features → requirements change → **Agile**.

Practice case: A bank wants a salary-calculation program for a fixed government formula that will not change. Requirements are clear and fixed, the project is medium-sized, and the technology is stable → **Waterfall**.

Practice case: A start-up wants a new food-delivery app and is not sure which features customers will like. Requirements are unclear and the project is innovative → **Agile**.

REMEMBER IT

One question decides it: "**Will the requirements change?**" No → Waterfall. Yes / not sure → Agile.

COMMON MISTAKE

Do not say one model is always "better". Each suits a different situation. In exams, justify your choice using the project's features (fixed vs changing requirements, size, customer involvement).

اردو میں سمجھیں

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔ ایجائل ماڈل لچکدار ہے، گاہک ہر اسپرنٹ کے بعد شامل ہوتا ہے، سافٹ ویئر بار بار حصوں میں ملتا ہے اور ٹیسٹنگ مسلسل ہوتی ہے۔ اگر ضروریات طے شدہ ہوں تو وائرفال اور اگر بدلتی رہیں تو ایجائل منتخب کریں۔

CHECK YOURSELF

1. In which model is the customer involved only at the start and the end?
2. Which model delivers software in frequent, partial deliveries?
3. A school wants an exam-result system with rules that never change. Which model?

Answers: 1. Waterfall. 2. Agile. 3. Waterfall (fixed, clear requirements).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Waterfall is:

- a. Linear — finish a phase before the next
- b. 2-week sprints
- c. Random order
- d. A gate

MCQ 2. Agile builds software in:

- a. Short sprints with feedback
- b. One giant leap only
- c. Hardware NAND
- d. OSI layer 1

MCQ 3. You cannot easily go back in:

- a. Waterfall
- b. Agile
- c. Both equally
- d. Neither

MCQ 4. Customer sees a working slice first in:

- a. Agile
- b. Pure waterfall
- c. K-maps
- d. Binary search

MCQ 5. A hospital records system with frozen law rules fits:

- a. Waterfall
- b. Only Agile memes
- c. No SDLC
- d. HCI pie charts

MCQ 6. Changing shop-app features weekly fits:

- a. Agile
- b. Waterfall only
- c. TCP
- d. Selection sort

Extra short questions

1. Define Waterfall and Agile in one line each.
2. Give one advantage of each.
3. Give one disadvantage of each.
4. When would you pick Waterfall?

Extra long question

1. Compare Waterfall vs Agile on five dimensions and attach one Sindh-college case to each model.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Waterfall: sequential phases, no going back. Agile: iterative sprints (about 2–4 weeks) with customer feedback.
2. Waterfall: clear documents and deadlines. Agile: change is cheap and the user sees value early.
3. Waterfall: late change is painful. Agile: needs an available customer and can look undocumented.
4. Stable, legally frozen requirements (e.g. a regulated records system).

Long

1. Dimensions: order, change, customer, documentation, risk. Case Waterfall: board exam-result system with fixed rules. Case Agile: college app whose features keep changing after student feedback.

1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model

★ Golden

LEARN IT

A **communication model** explains how data travels from one device to another through a network. Because sending data is complicated, the job is split into **layers**; each layer does one part of the job and passes the data to the next layer.

The **OSI (Open Systems Interconnection) model** was developed by **ISO** (International Organization for Standardization) in **1984**. It divides communication into **7 layers**. When you send data, it goes **down** the layers on your device (7 → 1), travels as bits over the cable or air, and goes **up** the layers on the receiving device (1 → 7).



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

REMEMBER IT

Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top to bottom you can use "All People Seem To Need Data Processing".)

OSI layers — top layers (7 to 4)

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII

5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)

OSI layers — bottom layers (3 to 1)

Layer	Function	Protocols / hardware
3. Network	Logical addressing (IP addresses) and routing packets across networks.	IP, ICMP (Ping), Routers
2. Data Link	Physical addressing (MAC addresses) and error detection in frames.	Ethernet, MAC, Switches
1. Physical	Transmits raw bits (0, 1) over cables or wireless media.	Cables, Hubs, Wi-Fi radio

TCP vs UDP at the Transport layer

	TCP	UDP
Main feature	Reliable — checks that all data arrives correctly and in order.	Fast — sends data without waiting for checks.
Use when	Accuracy matters.	Speed matters more than perfection.
Examples	Web browsing, emails	Video streaming, online gaming

WORKED EXAMPLE

Tracing an email through the OSI layers (sender side).

Layer 7 (Application): you press Send; your email program uses **SMTP**.

Layer 6 (Presentation): the text is encoded (ASCII) and may be encrypted with TLS.

Layer 5 (Session): a session is opened with the mail server.

Layer 4 (Transport): **TCP** breaks the data into segments and numbers them, so nothing is lost (email needs accuracy).

Layer 3 (Network): each packet gets the **IP address** of the destination; **routers** choose the path.

Layer 2 (Data Link): packets are put in **frames** with **MAC addresses**; a **switch** forwards them on the local network.

Layer 1 (Physical): the frames are sent as raw **bits** over the cable or Wi-Fi radio.

At the receiver, the same steps happen in reverse order (1 → 7).

COMMON MISTAKE

Do not mix up the two addresses: **IP address** = logical address = **Network layer (3)**; **MAC address** = physical address = **Data Link layer (2)**. Also, routers work at layer 3, switches at layer 2 and hubs at layer 1.

اردو میں سمجھیں

کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔ او ایس آئی (OSI) ماڈل 1984 میں آئی ایس او (ISO) نے بنایا اور اس کی سات تہیں (Layers) ہیں: فزیکل، ڈیٹا لنک، نیٹ ورک، ٹرانسپورٹ، سیشن، پریزنٹیشن اور ایپلی کیشن۔ ہر تہ کا الگ کام ہے، جیسے نیٹ ورک تہ IP ایڈریس اور روٹنگ سنبھالتی ہے۔ TCP درستگی کے لیے اور UDP رفتار کے لیے استعمال ہوتا ہے۔

EXAM TIP

A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two protocols/devices for each layer. MCQs often ask: "Which layer handles routing?" (Network), "Which layer does encryption?" (Presentation), "Who developed OSI and when?" (ISO, 1984), "Which protocol is used for video streaming?" (UDP).

CHECK YOURSELF

1. Which OSI layer is responsible for encryption and compression?
2. At which layer do switches and MAC addresses work?
3. Should online gaming use TCP or UDP? Why?

Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. OSI has:

- a. 7 layers
- b. 4 layers
- c. 2 layers
- d. 12 layers

MCQ 2. Layer 7 is:

- a. Application
- b. Physical
- c. Network
- d. Data link

MCQ 3. IP addressing lives at:

- a. Network (layer 3)
- b. Application
- c. Physical
- d. Session

MCQ 4. Cables and bits are:

- a. Physical
- b. Presentation
- c. Session
- d. Application

MCQ 5. End-to-end ports (TCP/UDP) are:

- a. Transport
- b. Network
- c. Physical
- d. Data link

MCQ 6. The model that explains “how data travels” is a:

- a. Communication model
- b. Sort algorithm
- c. K-map
- d. MVP

Extra short questions

1. Write the 7 OSI layers from top to bottom.
2. One job of Transport and of Network.
3. Why layer models?
4. Mnemonic you will actually write in the exam.

Extra long question

1. Name all 7 OSI layers with one job and one protocol/hardware example each. ★
Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Application, Presentation, Session, Transport, Network, Data Link, Physical.
2. Transport: reliability/ports (TCP/UDP). Network: routing and logical addressing (IP).
3. Each layer has a small job; vendors can change one layer without rewriting the rest.
4. Please Do Not Throw Sausage Pizza Away (bottom-up) or All People Seem To Need Data Processing (top-down).

Long

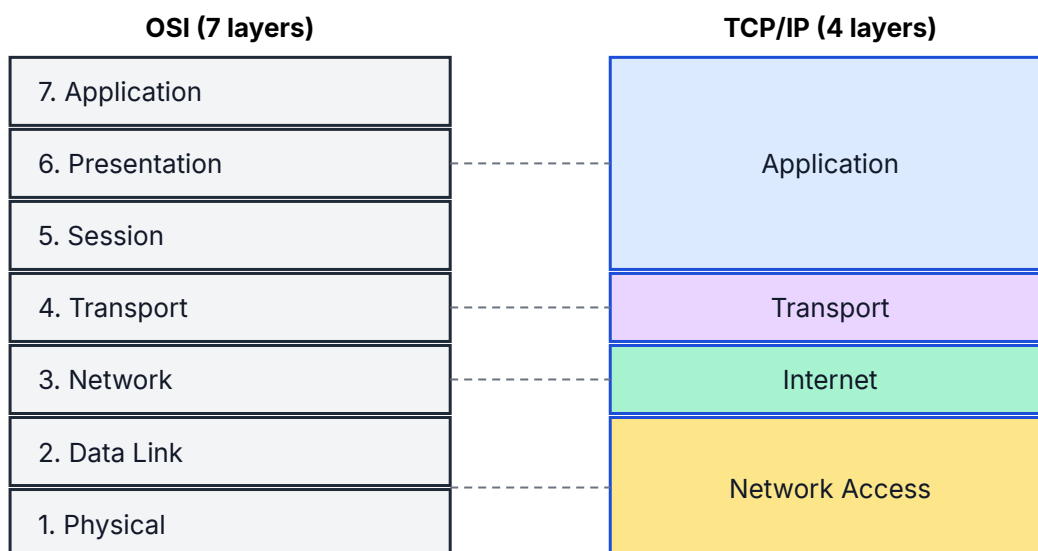
1. 7 App HTTP/DNS; 6 Pres encryption/JPEG; 5 Sess dialogs; 4 Trans TCP/UDP; 3 Net IP/routers; 2 DL frames/MAC/switches; 1 Phy cables/bits/hubs. Draw the stack.

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

LEARN IT

The OSI model is mainly a **reference** model used for teaching and design. The **TCP/IP model** is the **practical model used on the real Internet**. It condenses the 7 OSI layers into just **4 layers**.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Point	OSI model	TCP/IP model
Number of layers	7	4
Use	Reference / teaching model	Practical model used on the real Internet
Top layers	Separate Application, Presentation and Session layers	All three combined into one Application layer
Bottom layers	Separate Data Link and Physical layers	Combined into one Network Access layer

WORKED EXAMPLE

Where does each protocol/device go in TCP/IP?

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

REMEMBER IT

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total $3+1+1+2 = 7$.

COMMON MISTAKE

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

اردو میں سمجھیں

ٹی سی پی/آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

CHECK YOURSELF

1. How many layers does the TCP/IP model have?
2. Which TCP/IP layer corresponds to OSI layers 2 and 1?
3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. OSI has:

- a. 7 layers
- b. 4 layers
- c. 2 layers
- d. 12 layers

MCQ 2. Layer 7 is:

- a. Application
- b. Physical
- c. Network
- d. Data link

MCQ 3. IP addressing lives at:

- a. Network (layer 3)
- b. Application
- c. Physical
- d. Session

MCQ 4. Cables and bits are:

- a. Physical
- b. Presentation
- c. Session
- d. Application

MCQ 5. End-to-end ports (TCP/UDP) are:

- a. Transport
- b. Network
- c. Physical
- d. Data link

MCQ 6. The model that explains “how data travels” is a:

- a. Communication model
- b. Sort algorithm
- c. K-map
- d. MVP

Extra short questions

1. Write the 7 OSI layers from top to bottom.
2. One job of Transport and of Network.
3. Why layer models?
4. Mnemonic you will actually write in the exam.

Extra long question

1. Name all 7 OSI layers with one job and one protocol/hardware example each. ★
Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Application, Presentation, Session, Transport, Network, Data Link, Physical.
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Long

1. 7 App HTTP/DNS; 6 Pres encryption/JPEG; 5 Sess dialogs; 4 Trans TCP/UDP; 3 Net IP/routers; 2 DL frames/MAC/switches; 1 Phy cables/bits/hubs. Draw the stack.

Computer Systems

12 single-page lessons · 7 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (') inverts its single input (unary).
- A truth table lists all 2^n input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Key terms

Term	Meaning	اردو
Discrete quantity	A countable quantity with separate, fixed values.	منفصل مقدار
Continuous quantity	A quantity that can take any value within a range.	مسلسل مقدار
Bit	A binary digit, 0 or 1.	بائنری ہندسہ
Digital system	An electronic system that uses only the values 0 and 1.	ڈیجیٹل نظام

Analog signal	A continuous signal that changes smoothly, like a sine wave.	اینالگ سگنل (مسلسل اشارہ)
Digital signal	A signal with only two levels, HIGH and LOW (square wave).	ڈیجیٹل سگنل
Noise	Unwanted interference that disturbs a signal.	شور (مداخلت)
Boolean algebra	Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole.	بولین الجبرا
Truth table	A table of all input combinations and their outputs.	ٹرو تھ ٹیبل (صداقت جدول)
Logic gate	An electronic circuit that performs a Boolean operation.	منطقی گیٹ
Universal gate	A gate (NAND or NOR) from which any circuit can be built.	عالمگیر گیٹ
Unary operation	An operation with only one operand, e.g. NOT.	یک عددی عمل

CHAPTER 1 EXAM • CLOSED BOOK

Computer Systems

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 12 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. 15 + 30 + 30 = 75. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

- Which of the following is a continuous quantity?
 - Number of students in a class
 - Number of cars in a parking lot
 - Temperature of water
 - Number of books on a shelf
- In a digital system, the value 1 represents:
 - OFF
 - LOW
 - FALSE
 - ON / HIGH / TRUE

3. A square wave represents a:
- a. Digital signal
 - b. Analog signal
 - c. Sine wave
 - d. Noise signal
4. Boolean algebra was developed by:
- a. Charles Babbage
 - b. George Boole
 - c. Alan Turing
 - d. Maurice Karnaugh
5. How many rows does a truth table with 4 inputs have?
- a. 4
 - b. 8
 - c. 16
 - d. 32
6. Which gate gives output 1 only when all inputs are 1?
- a. OR
 - b. AND
 - c. NOR
 - d. XOR
7. Which pair of gates are called universal gates?
- a. AND and OR
 - b. XOR and XNOR
 - c. NOT and AND
 - d. NAND and NOR
8. What is the output of a NOR gate when $A = 0$ and $B = 0$?
- a. 1
 - b. 0
 - c. Unknown
 - d. Depends on the gate size
9. An XOR gate gives output 1 when:
- a. Both inputs are 1
 - b. Both inputs are 0
 - c. The inputs are different
 - d. The inputs are the same
10. After parentheses, which operation has the highest precedence?
- a. OR
 - b. NOT
 - c. AND
 - d. XOR
11. The minterm for the row $A = 1, B = 0$ is:
- a. AB'
 - b. $A'B$
 - c. $A + B'$
 - d. $A' + B$
12. The columns of a 3-variable K-map (BC) are written in the order:
- a. 00, 01, 10, 11
 - b. 00, 10, 01, 11
 - c. 11, 10, 01, 00
 - d. 00, 01, 11, 10

3. In black box testing, the tester:
- a. Reads the source code line by line
 - b. Does not see the code and tests inputs/outputs
 - c. Is always the developer
 - d. Tests only the internal structure
4. Which OSI layer handles logical (IP) addressing and routing?
- a. Data Link
 - b. Transport
 - c. Network
 - d. Session
5. The TCP/IP Network Access layer corresponds to which OSI layers?
- a. 7, 6 and 5
 - b. 4
 - c. 3
 - d. 2 and 1

Section B — Short questions (10 × 3 = 30)

1. Define discrete and continuous quantities. Give two examples of each.
2. State four advantages of digital systems.
3. Differentiate between analog and digital signals (any three points).
4. Why is NOT called a unary operation? Write its truth table.
5. How many rows are needed in a truth table for 3 inputs? List all the input combinations.
6. Why are NAND and NOR called universal gates? Show how a NAND gate can be used as a NOT gate.
7. A 2-input truth table has $Y = 1$ only for rows $A=0, B=1$ and $A=1, B=0$. Write the SOP expression and name the gate.
8. Name the deliverable of each SDLC phase.
9. Differentiate between black box testing and white box testing.
10. When should TCP be used and when should UDP be used? Give one example of each.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Draw the symbol, write the Boolean expression and give the truth table for each of these gates: AND, OR, NOT, NAND, NOR, XOR and XNOR.
2. A function is given as $F(A, B, C) = \sum m(1, 3, 5, 6, 7)$. (a) Make its truth table. (b) Simplify it using a 3-variable K-map, explaining each group. (c) Draw the logic diagram of the simplified expression.
3. For $Y = A \cdot B' + C$: (a) state the order in which the operations are done; (b) make the complete truth table; (c) draw the logic diagram.
4. Explain the six phases of the Software Development Life Cycle with the activities and deliverable of each phase. Use the house-building analogy.

5. Compare the Waterfall and Agile models under five headings. Explain why the AI-Noor Library Management System used Waterfall and the AI-Noor Student Mobile Application used Agile.
6. Explain the seven layers of the OSI model with the function and one protocol/device for each layer. Then show how they map onto the four layers of the TCP/IP model.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

- 1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems.** Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.
- 2. 1.1.3 Analog and Digital Signals.** Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).
- 3. 1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables.** Answers: 1. George Boole. 2. 1 (because + means OR).
- 4. 1.1.7 Basic Logic Gates (AND, OR, NOT).** Answers: 1. 0. 2. Inverter. 3. Only $A = 1, B = 1$.
- 5. 1.1.7 Universal & Advanced Logic Gates.** Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.
- 6. 1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams.** Answers: 1. AND. 2. $AB'C$. 3. $A + B'$.
- 7. 1.1.13 Karnaugh Maps (K-Maps) Simplification.** Answers: 1. $BC = 11$. 2. Cells 2 and 3 are the row $A = 1$; B changes $\rightarrow Y = A$. 3. The whole row $A = 1$ is a group of 4 $\rightarrow Y = A$.
- 8. LogiSim Simulation Tool Guide.** Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.
- 9. 1.2.1 – 1.2.3 Software Development Life Cycle (SDLC).** Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.
- 10. 1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies.** Answers: 1. Because each phase must be completed before the next begins, in a fixed order. 2. Because its requirements were 100% fixed before coding began.
- 11. 1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model.** Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.
- 12. 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model.** Answers: 1. Four. 2. Network Access. 3. Application.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. c | 4. b | 7. d | 10. b | 13. b |
| 2. d | 5. c | 8. a | 11. a | 14. c |
| 3. a | 6. b | 9. c | 12. d | 15. d |

Short questions — model answers

- Discrete quantities are separate, countable items with specific values (number of students in a classroom, number of cars in a parking lot). Continuous quantities can take any value within a range and flow without breaks (temperature of water, speed of a moving car).
- Reliable (less affected by noise and interference); accurate (no loss of quality when copying); easy to design (only two states, 0 and 1); programmable (easily controlled by software).
- Nature: analog is a continuous flow; digital has discrete values (bits, HIGH/LOW). Shape: analog is a sine wave; digital is a square wave. Reliability: analog is low (affected by

noise); digital is high (resistant to noise). Examples: human voice vs computer data.

4. NOT works on only one operand (one input), so it is unary. Truth table: $A = 0 \rightarrow Y = 1$; $A = 1 \rightarrow Y = 0$.
5. $2^3 = 8$ rows: 000, 001, 010, 011, 100, 101, 110, 111.
6. Any logic circuit can be built using only NAND gates, or only NOR gates. NAND as NOT: join both inputs and feed A: $Y = (A \cdot A)' = A'$. Check: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.
7. Minterms: $A'B$ (row 01) and AB' (row 10). $Y = A'B + AB'$. This is the XOR gate, $Y = A \oplus B$.
8. Requirement Analysis $\rightarrow$ SRS document; System Design $\rightarrow$ Design Document; Implementation $\rightarrow$ Working Source Code; Testing $\rightarrow$ Test Reports / bug-free software; Deployment $\rightarrow$ Live Software; Maintenance $\rightarrow$ Patches / Updated Versions.
9. Black box: tester does not see the code, tests functionality (inputs/outputs), done by dedicated testers. White box: tester can see the code, tests internal logic/structure, done by developers.
10. Use TCP when accuracy matters, e.g. web browsing or email. Use UDP when speed matters more than perfection, e.g. video streaming or online gaming.

Long questions — model answers

1. Symbols as drawn in 1.1.7 (AND = D shape; OR = curved back, pointed front; NOT = triangle + bubble; NAND = AND + bubble; NOR = OR + bubble; XOR = OR with extra back curve; XNOR = XOR + bubble). Expressions: AND $Y = A \cdot B$; OR $Y = A + B$; NOT $Y = A'$; NAND $Y = (A \cdot B)'$; NOR $Y = (A + B)'$; XOR $Y = A \oplus B$; XNOR $Y = (A \oplus B)'$. Outputs for inputs 00, 01, 10, 11: AND 0,0,0,1; OR 0,1,1,1; NAND 1,1,1,0; NOR 1,0,0,0; XOR 0,1,1,0; XNOR 1,0,0,1. NOT: $0 \rightarrow 1$, $1 \rightarrow 0$.
2. (a) Truth table ($ABC \rightarrow F$): 000 $\rightarrow$ 0, 001 $\rightarrow$ 1, 010 $\rightarrow$ 0, 011 $\rightarrow$ 1, 100 $\rightarrow$ 0, 101 $\rightarrow$ 1, 110 $\rightarrow$ 1, 111 $\rightarrow$ 1.
(b) K-map:

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	1	0
A = 1	0	1	1	1

Group 1 (size 4): the two middle columns, m_1, m_3, m_5, m_7 . A changes and B changes; C = 1 stays $\rightarrow$ **C**. Group 2 (size 2): m_7 and m_6 in row A = 1. C changes; A = 1, B = 1 stay $\rightarrow$ **AB** (m_7 is shared — overlapping is allowed). **F = C + AB**. Check: C + AB is 1 for rows 1, 3, 5, 7 (C = 1) and 6, 7 (AB = 1) = {1, 3, 5, 6, 7}. ✓

(c) Logic diagram: an AND gate with inputs A and B gives AB; an OR gate takes AB and C and gives F.

3. (a) First NOT (B'), then AND ($A \cdot B'$), then OR ($+ C$).

(b) Truth table:

A	B	C	B'	A·B'	Y
---	---	---	----	------	---

0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

(c) Logic diagram: input B goes through a NOT gate to give B'. A and B' go into an AND gate to give A·B'. A·B' and C go into an OR gate to give Y. (3 gates.)

4. Requirement Analysis: meet clients, understand needs, list features; most important phase; deliverable SRS (house: asking the family what rooms they need). System Design: high-level design (architecture), low-level design (database, UI screens), flowcharts; deliverable Design Document (architect draws blueprints). Implementation: write code in Python, Java etc. from the design; deliverable Working Source Code (builders lay bricks). Testing: find and fix bugs using black box and white box testing; deliverable Test Reports / bug-free software (inspector checks the house). Deployment: release to end-users by direct, phased or pilot release; deliverable Live Software (family moves in). Maintenance: support, updates, fixing post-release issues; deliverable Patches / Updated Versions (repairs and extensions).
5. Comparison: Flexibility — Waterfall rigid, Agile very flexible. Customer involvement — Waterfall only at start and end, Agile continuous after every sprint. Delivery — Waterfall single delivery at the end, Agile frequent partial deliveries. Testing — Waterfall late in a dedicated phase, Agile continuous in every sprint. Documentation — Waterfall heavy, Agile light. The Al-Noor Library system used Waterfall because its requirements were 100% fixed before coding began, so a simple, well-documented linear process fitted. The Al-Noor Student App used Agile because teachers, parents and students kept requesting new features, so short sprints with continuous feedback were needed.
6. OSI (ISO, 1984), top to bottom: 7 Application — network services to user applications (HTTP, HTTPS, FTP, SMTP, DNS); 6 Presentation — translation, encryption, compression (SSL/TLS, JPEG, MP3, ASCII); 5 Session — opens, maintains, closes sessions (NetBIOS, RPC); 4 Transport — reliable end-to-end delivery, error checking, port numbers (TCP, UDP); 3 Network — logical IP addressing and routing (IP, ICMP, routers); 2 Data Link — physical MAC addressing and error detection in frames (Ethernet, switches); 1 Physical — raw bits over cables/wireless (cables, hubs, Wi-Fi radio). Mnemonic bottom to top: "Please Do Not Throw Sausage Pizza Away". TCP/IP mapping: Application = OSI 7, 6, 5; Transport = 4; Internet = 3; Network Access = 2, 1.

Self-assessment: I can...

- ☐ ...tell discrete from continuous quantities and analog from digital signals, and draw both waves.
- ☐ ...evaluate AND, OR and NOT expressions and follow the order of precedence.
- ☐ ...draw all seven gate symbols and write their truth tables from memory.
- ☐ ...write the SOP (minterms) and POS (maxterms) from a truth table.
- ☐ ...simplify expressions with 2- and 3-variable K-maps, including overlap and wrap-around groups.
- ☐ ...explain the six SDLC phases and choose between Waterfall and Agile for a project.
- ☐ ...list four advantages of digital systems.
- ☐ ...build a complete truth table for 2 or 3 inputs without missing a row.
- ☐ ...explain why NAND and NOR are universal gates.
- ☐ ...draw a logic diagram from an expression.
- ☐ ...build and test a simple circuit in LogiSim and read the wire colours.
- ☐ ...name the 7 OSI layers in order, state their functions, and map them to the 4 TCP/IP layers.